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Lubricated trailer ball and receiver assembly

Abstract

A system is disclosed herein including a tow vehicle including and a trailer. The tow vehicle includes a bed having a first surface and an opposing second surface and a receiver assembly coupled to the bed, the receiver assembly extending through the first surface and the second surface. The receiver assembly includes a hollow cylinder having a first end and a second end, the first end of the hollow cylinder coupled to the bed of the tow vehicle and a ball receiver having a semi-spherical shape, the ball receiver coupled to the second end of the hollow cylinder. The trailer includes a deck, a neck coupled to the deck and a trailer coupler coupled to the neck. The trailer coupler includes a trailer ball configured to engage the receiver assembly of the tow vehicle.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a divisional of U.S. patent application Ser. No. 18/112,044, entitled “Lubricated Trailer Ball and Receiver Assembly,” filed Feb. 21, 2023 which claims priority to, and the benefit of, U.S. Provisional Patent Application No. 63/313,154, filed Feb. 23, 2022, and titled “Lubricated Trailer Ball and Receiver Assembly,” both of which are incorporated by reference herein in their entirety for all purposes.

FIELD

(1) The present disclosure generally relates trailer hitches, and more specifically, to trailer hitch ball and receiver assemblies.

BACKGROUND

(2) Generally, tow vehicles, including pickup truck vehicles and semi tractors, and trailers can be identified as class 2 through 8 type vehicles, whereby semi tractors can be identified as class 8. These different class vehicles share the same similarity of being capable of pulling various types of trailers. Pulling such trailers requires a device known as a hitch. Some common examples of hitches include bumper hitch, receiver and ball platform, gooseneck hitch, and fifth wheel hitch. The conventional hitch assembly of these class vehicles and/or the trailers they tow has seen only minor design revisions over many years, even despite vehicle engine horse power, torque, and Gross Vehicle Weight Rating (GVWR) having increased through the years.

(3) Fifth wheel hitches use jaws that latch onto a protruded “stem” of a semi-trailer. The “stem” is embedded within a thick plate of sheet steel that is attached under the deck, or floor, of the semi-trailer. The stem, also referred to as a King Pin, is a machined round rod having a machined center groove that the 5th wheel jaws to engages. Over time, wear and usage causes deterioration in the tolerances of the jaws. Loose jaws may cause a noticeable shock load to the tow vehicle which can be a safety concern. The fifth wheel jaws are typically tighten by tightening a threaded rod, that if tightened incorrectly, may allow the stem, or King Pin, to disengage from the jaws of the fifth wheel hitch.

(4) Typically, lower class tow vehicles (e.g., class 2) can tow lesser rated trailers with either a bumper hitch, a receiver and ball hitch, a goose-neck hitch, or miniature fifth wheel hitch, such as used in travel trailers. Typically, industry standards can refer to the goose-neck design as a ball and coupler with the ball installed on the tow vehicle and the coupler installed on the trailer. The conventional ball diameters are generally between 1 7/8 inches and 3 inches, with 2 5/16 inches (about 5.87 centimeters) being common, and the coupler installed on the trailer has an orifice sized to engage the ball. Typically, the coupler will be equipped with a latching mechanism intended to retain the coupler from separating from the ball installed on the tow vehicle. Further, the trailer frame coupler is typically equipped with safety chains or cables, as required by the US Department of Transportation (US-DOT).

(5) The conventional tow ball (e.g., 2 5/16 inches (5.87 centimeters)) and coupler have a “dry” fit connection with respect to each other. As a result, each movement of the tow vehicle or trailer is subject to a coefficient of friction creating abrasion on the ball and coupler via the dry frictional forces and surface tensions between the ball and coupler. Specifically, it has been observed that through time this friction and abrasion reduces the ball diameter and enlarges the receiving coupler orifice, thereby increasing the risk of the two components disconnecting, which could have serious and/or fatal consequences on the road.

(6) For trailer frames with embedded couplers, the coupler extends adjustable dimensions below the trailer neck frame. That extension then becomes a lever arm, resulting in a trailer twisting moment under various conditions. An increasing percentage of trailers are constructed with Aluminum. Aluminum is especially known to work harden and fatigue from bending, or rigidity movement, resulting in failure. There again, the ball and coupler present problems with material fatigue and/or failure. As the coupler rotates, friction restricts rotation, the coupler cogs on the ball, releases from the ball, and creates a shock load to both the tow vehicle and the trailer.

SUMMARY

- (7) Disclosed herein is a trailer ball and receiver assembly, including a trailer ball assembly coupled to a trailer, the trailer ball assembly including a trailer ball and a ball receiver assembly coupled to a tow vehicle, the ball receiver assembly including a semi-spherical socket configured to receive the trailer ball assembly.
- (8) In various embodiments, the receiver assembly further includes a lubricant channel formed in a bottom of the semi-spherical socket and extending through a bottom of the ball receiver assembly and a plug configured to be inserted into and removed from the lubricant channel, the plug further configured to prevent liquid from passing through the lubricant channel when inserted into the lubricant channel.
- (9) In various embodiments, the receiver assembly further includes a cylinder having a first end and opposing second end and a hole formed from the first end to the second end, wherein the second end of the cylinder is coupled to the ball receiver assembly so that the semi-spherical socket opens toward the hole.
- (10) In various embodiments, the semi-spherical socket has a diameter of about 5.5 inches to about 7.5 inches. In various embodiments, the trailer ball has a diameter of about 5 inches to about 7 inches. In various embodiments, the trailer ball and receiver assembly further includes a ball mount configured to receive the trailer ball and secure the trailer ball in place and a trailer coupler adapter having a first end and a second end, the first end coupled to the ball mount and the second end coupled to the trailer. In various embodiments, the trailer ball and receiver assembly further includes a lubricant disposed within the semi-spherical socket and between the semi-spherical socket and the trailer ball.
- (11) Also disclosed herein is a system including a tow vehicle including and a trailer. The tow vehicle includes a bed having a first surface and an opposing second surface and a receiver assembly coupled to the bed, the receiver assembly extending through the first surface and the second surface. The receiver assembly includes a hollow cylinder having a first end and a second end, the first end of the hollow cylinder coupled to the bed of the tow vehicle and a ball receiver having a semi-spherical shape, the ball receiver coupled to the second end of the hollow cylinder. The trailer includes a deck, a neck coupled to the deck, and a trailer coupler coupled to the neck. The trailer coupler includes a trailer ball configured to engage the receiver assembly of the tow vehicle.
- (12) In various embodiments, the system further includes a lubricant disposed in the receiver assembly, the lubricant disposed between the trailer ball and the ball receiver. In various embodiments, the receiver assembly further includes a channel formed in and through a bottom of the ball receiver, the channel configured to drain the lubricant from the ball receiver and a removable plug configured to be inserted into the channel, the removable plug preventing the lubricant from draining from the ball receiver.
- (13) In various embodiments, the trailer coupler further includes a trailer coupler adapter, a ball mount configured to receive the trailer ball, and a damper disposed between the ball mount and the trailer coupler adapter. In various embodiments, the system further includes a plurality of holes formed along a length of the trailer coupler adapter, the plurality of holes configured to secure the trailer coupler adapter to the neck of the trailer.
- (14) In various embodiments, the trailer ball has a first diameter, the first diameter being about 5 inches to about 7 inches. In various embodiments, the semi-spherical shape of the ball receiver has a second diameter, the second diameter being about 1/16 inch larger than the first diameter.
- (15) Also disclosed herein is a towing system including a tow vehicle having a bed, a trailer, and a lubricated ball and receiver assembly coupling the trailer to the tow vehicle. The lubricated ball and receiver assembly includes a receiver assembly having a first end and a second end, the first end coupled to the bed of the tow vehicle, the second end extending downward from the bed of the tow vehicle, the second end of the receiver assembly ending in a semi-spherical ball receiver, a drain

formed in the semi-spherical ball receiver and through the second end of the receiver assembly, a lubricant disposed in the semi-spherical ball receiver, a trailer ball sized to pass through the receiver assembly and into the semi-spherical ball receiver, and a trailer coupler including a first vertical member having a first end and a second end, the first end of the first vertical member coupled to the trailer, and the second end of the first vertical member coupled to the trailer ball.

(16) In various embodiments, the towing system further includes a plug configured to be inserted into the drain, the plug preventing the lubricant from exiting the receiver assembly when inserted. In various embodiments, the trailer ball has a first diameter and the semi-spherical ball receiver has a second diameter, the second diameter being larger than the first diameter. In various embodiments, the first diameter is about 5 inches to about 7 inches and the second diameter is about 1/16 inches to about 3/16 inches larger than the first diameter.

(17) In various embodiments, the trailer coupler further includes a second vertical member having a first end and a second end and a damper disposed between the first vertical member and the second vertical member. In various embodiments, the receiver assembly extends about 3 inches to about 18 inches below the bed of the tow vehicle.

(18) The foregoing features and elements may be combined in any combination, without exclusivity, unless expressly indicated herein otherwise. These features and elements as well as the operation of the disclosed embodiments will become more apparent in light of the following description and accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The subject matter of the present disclosure is particularly pointed out and distinctly claimed in the concluding portion of the specification. A more complete understanding of the present disclosure, however, may best be obtained by referring to the following detailed description and claims in connection with the following drawings. While the drawings illustrate various embodiments employing the principles described herein, the drawings do not limit the scope of the claims.

(2) FIG. 1 illustrates a truck and trailer having a trailer hitch ball and receiver assembly, in accordance with various embodiments.

(3) FIG. 2 illustrates a cross-sectional side view of a trailer hitch ball and receiver assembly, in accordance with various embodiments.

(4) FIG. 3 illustrates a cross-sectional side view of aspects of a trailer hitch ball and receiver assembly, in accordance with various embodiments.

(5) FIG. 4 illustrates a cross-sectional side view of aspects of a trailer hitch ball and receiver assembly, in accordance with various embodiments.

(6) FIGS. 5A, 5B, 5C, 5D, and 5E illustrate aspects of a trailer hitch receiver assembly, in accordance with various embodiments.

(7) FIGS. 6A, 6B, 6C, and 6D illustrate aspects of a trailer hitch ball assembly, in accordance with various embodiments.

DETAILED DESCRIPTION

(8) The following detailed description of various embodiments herein makes reference to the accompanying drawings, which show various embodiments by way of illustration. While these various embodiments are described in sufficient detail to enable those skilled in the art to practice the disclosure, it should be understood that other embodiments may be realized and that changes may be made without departing from the scope of the disclosure. Thus, the detailed description herein is presented for purposes of illustration only and not of limitation. While these exemplary embodiments are described in sufficient detail to enable those skilled in the art to practice the

invention, it should be understood that other embodiments may be realized and that logical, chemical and mechanical changes may be made without departing from the spirit and scope of the invention. For example, the steps recited in any of the method or process descriptions may be executed in any order and are not necessarily limited to the order presented. Furthermore, any reference to singular includes plural embodiments, and any reference to more than one component or step may include a singular embodiment or step. Also, any reference to attached, fixed, connected, or the like may include permanent, removable, temporary, partial, full or any other possible attachment option. Additionally, any reference to without contact (or similar phrases) may also include reduced contact or minimal contact. It should also be understood that unless specifically stated otherwise, references to “a,” “an” or “the” may include one or more than one and that reference to an item in the singular may also include the item in the plural. Further, all ranges may include upper and lower values and all ranges and ratio limits disclosed herein may be combined.

(9) Disclosed herein is a trailer ball and receiver assembly for use with tow vehicles and trailers. The trailer ball and receiver may replace conventional hitch balls and couplers currently in use. In various embodiments, a receiver assembly is coupled to a tow vehicle, and more specifically, below a bed of the tow vehicle. In various embodiments, a cross member is coupled to the tow vehicle, below the bed of the tow vehicle, and extending from one side of the tow vehicle to the other. The cross member adds rigidity to the frame of the tow vehicle as well as support for the receiver assembly. In various embodiments, the receiver assembly includes a semi-circular receiver and a cylindrical component. In various embodiments, a top portion of the cylindrical component may be coupled the bed of the tow vehicle with the cylinder extending downward from the bed of the tow vehicle. In various embodiments, the semi-circular receiver may be coupled a bottom portion of the cylindrical component. In various embodiments, the receiver assembly, and more specifically, the semi-circular receiver is configured to receive and secure a trailer ball, or simply, ball.

(10) In various embodiments, the trailer ball, or ball, may have a diameter of about 4 inches (about 10.2 centimeters) to about 8 inches (about 20.3 centimeters). In various embodiments, the ball may be coupled to a neck of a trailer. In various embodiments, the trailer may be a gooseneck trailer or a fifth wheel trailer. In various embodiments, an adapter, or ball mount, may be used to couple the ball to the neck of the trailer. In various embodiments, the ball and ball mount may be coupled to the tow vehicle by lowering the ball into the receiver assembly, and more specifically, into the cylindrical component to engage the semi-circular receiver.

(11) In various embodiments, the trailer ball and receiver assembly may be lubricated. The lubricant may reduce the friction between the ball and the semi-circular receiver. This may result in less cogging, or catching, between the ball and the receiver, reducing force transfer from the tow vehicle to the trailer. This decreases stresses and wear on the trailer and on the tow vehicle, improving the functional life of both the trailer and the tow vehicle.

(12) The trailer ball and receiver assembly disclosed herein may be retrofitted to existing trailers and tow vehicles, respectively. As disclosed herein the, the trailer ball and receiver assembly does not allow the trailer ball to disengage from the receiver assembly, thereby removing common safety concerns with other types of hitches (e.g., fifth wheels). Additionally, the trailer ball and receiver assembly may improve the ride of the tow vehicle, the handling of the tow vehicle, and wear on both the tow vehicle and the trailer. In various embodiments, the tow vehicle may see improved fuel efficiency using the trailer ball and receiver assembly disclosed herein as compared with current hitch ball and coupler designs. Some of the improved fuel efficiency may be achieved through a reduction in weight as the ball and receiver assembly disclosed herein may be lighter than conventional fifth wheel hitches. In various embodiments, the trailer may experience less twisting and/or torquing when using the trailer ball and receiver assembly as compared to as compared to existing hitch ball and coupler assemblies, resulting in an increased useable life of the trailer. In various embodiments, the reduced twisting, or torquing, of the trailer may result in

decreased work hardening of the trailer structure. Additionally, the trailer ball and receiver assembly disclosed herein eliminates the use of expensive pads that are currently used to reduce abrasion and wear in fifth wheel hitches.

(13) Referring now to FIG. 1, a system **100** including a tow vehicle **102** and a trailer **104** are illustrated, in accordance with various embodiments. In various embodiments, tow vehicle **102** may be a pickup truck, a semi-tractor, a tractor, a cab over truck, or a terminal truck (e.g., yard truck, yard goat, etc.), among other types of vehicles used for towing. Tow vehicle **102** includes a cab **106** and a bed **108** located behind cab **106** (e.g., in the negative y-direction). Cab **106** may accommodate a driver and one or more passengers. Bed **108** has a top surface **108a** and bottom surface **108b**. Bed **108** may be used for hauling cargo (e.g., on top surface **108a**) and/or coupling trailer **104** to tow vehicle **102**. System **100** further includes a conventional trailer ball **110** coupled to bed **108** (e.g., on top surface **108a**) and a trailer ball and receiver system **112**.

(14) Trailer ball and receiver system **112** includes a receiver assembly **144** and a trailer ball **140**. In various embodiments, receiver assembly **144** may be coupled to bed **108** (e.g., through top surface **108a** and bottom surface **108b**). In various embodiments, trailer ball **110** may be a conventional 2 5/16 inches (5.87 centimeters) trailer ball coupled to top surface **108a** of bed **108**. That is, trailer ball **110** may extend above bed **108** (e.g., in the positive z-direction and over top surface **108a**). In various embodiments, receiver assembly **144** may be positioned behind trailer ball (e.g., in the negative y-direction). Receiver assembly **144** may extend below bed **108** (e.g., in the negative z-direction or toward a road surface). In various embodiments, trailer ball and receiver system **112** may further include a cross member **146** that extends from one side of tow vehicle **102** to the other side (e.g., along the x-axis). Receiver assembly **144** may extend into and/or through cross member **146**. Cross member **146** may provide support for receiver assembly **144** and/or rigidity to tow vehicle **102**.

(15) In various embodiments, receiver assembly **144** may interfere with an axle housing of tow vehicle **102**. In such embodiments, the axle housing may be modified to accommodate installation of receiver assembly **144**. Clearance between receiver housing **144** and the axle housing may be reduced, and in various embodiments eliminated, by modifying the axle housing to accommodate receiver assembly **144**. In various embodiments, receiver assembly **144** may rest on the axle housing. In various embodiments, receiver assembly **144** may be attached to axle housing. In various embodiments, receiver assembly **144** may be cast into the axle housing at the time of production. By so doing, the added weight of receiver housing **144**, and more generally trailer ball and receiver system **112**, is supported by the axle housing and may add little to no load compression to the suspension, frame rails, springs, and other components of tow vehicle **102**.

(16) In various embodiments, tow vehicle **102** may include only receiver assembly **144**. In various embodiments, tow vehicle **102** may omit the conventional trailer ball **110**. It should be appreciated that while a conventional trailer ball **110** (e.g., a gooseneck hitch) is illustrated in FIG. 1, tow vehicle **102** may include a bumper hitch, a fifth wheel hitch, and/or a conventional receiver and ball style hitch. It should further be appreciated that receiver assembly **144** may be installed in, or on, tow vehicle **102** either alone or in combination with any other hitch such as a bumper hitch, a gooseneck hitch (e.g., trailer ball **110**), a fifth wheel hitch, and/or a conventional receiver and ball style hitch, among others. Furthermore, in various embodiments, receiver assembly **144** may be installed in, or on, a dolly to allow towing a second trailer. Therefore, it should be understood that tow vehicle **102** illustrated in FIG. 1 is not meant to be limiting but for aiding in the discussion below.

(17) As illustrated, trailer **104** is connected to tow vehicle **102** using trailer ball and receiver system **112**. In various embodiments, trailer **104** may be a gooseneck trailer, a fifth wheel trailer, or a bumper pull trailer. Trailer **104** includes a coupler hitch **120**, one or more safety chains **122**, a nose plate **124**, a neck **126**, a frame **128**, a deck **130**, and one or more drop log jacks **132**. Deck **130** is coupled to frame **128** providing a surface on which to secure materials for transport. In various

embodiments, trailer **104** may be enclosed such as, for example, a box trailer, a refrigerator trailer, a livestock or animal trailer, or a cargo trailer, among others.

(18) Neck **126** is coupled to a forward portion (e.g., toward the positive y-direction) of frame **128** and is configured to connect trailer **104** to tow vehicle **102** while allowing deck **130** to be level at a desired height (spacing along the z-axis relative to a road surface). In various embodiments, the one or more drop log jacks **132** may be coupled to the forward portion of frame **128** below (e.g., disposed farther in a negative-z direction or nearer the road surface) neck **126**. Drop log jacks **132** provide support for trailer **104** when trailer **104** is not coupled to tow vehicle **102**. In various embodiments, different aspects of trailer **104**, including the aspects discussed herein, may be telescoping or rotating to provide more or less height as needed. Nose plate **124** is coupled to a forward portion of neck **126** and, when being towed, is disposed over bed **108** of tow vehicle **102**. Coupler hitch **120** and safety chains **122** are coupled to neck **126**, below (e.g., disposed farther in a negative-z direction or nearer the road surface) nose plate **124**.

(19) Coupler hitch **120** includes a trailer ball **140** coupled to neck **126** by a vertical support **142**. Receiver assembly **144** is further configured a vertical support **143** coupled to trailer ball **140**. That is, in contrast with conventional towing configurations in which trailer ball **110** is coupled to tow vehicle **102** and a coupler is connected to trailer **104** to engage trailer ball **110**, system **100** is designed so that trailer ball **140** is coupled to trailer **104** (e.g., coupler hitch **120**) and receiver assembly **144** is coupled to tow vehicle **102**. Additionally, trailer ball **140** may be larger than conventional trailer balls resulting in an increased towing capacity, as will be described in further detail below. This configuration provides an improved towing experience that will become apparent to the reader as trailer ball and receiver system **112** is described in further detail below. It should be appreciated that tow vehicle **102** and trailer **104** as illustrated in FIG. 1 are not intended to be limiting as to the application of the trailer ball and receiver system **112** disclosed herein, but are used for discussion purposes only. As such, any suitable tow vehicle **102** may be used to tow any suitable trailer **104** using trailer ball and receiver system **112** disclosed herein.

(20) As disclosed herein, trailer ball and receiver system **112** provides lateral (e.g., along the x-axis) and longitudinal (e.g., along the y-axis) support. The additional support provided by trailer ball and receiver system **112** over conventional hitches increases the safety of tow vehicle **102** and trailer **104** while in use. Furthermore, trailer ball and receiver system **112** lowers the center line of the pulling force of tow vehicle (i.e., below bed **108**) towards the center line of the wheels of trailer **104** (e.g., line **150**). Lowering the center line of the pulling force (e.g., line **152**) allows tow vehicle **102** to pull the nose of trailer **104** (e.g., nose plate **124**) more parallel to the road (e.g., the y-axis) as compared to a typical fifth wheel hitch (e.g., line **154**) which applies an upward force (e.g., in the positive z-direction) to the nose of trailer **104**. There is an angle α_1 between line **152** and line **150** (e.g., force line of trailer ball and receiver system **112**) and an angle α_2 between line **154** and line **150** (e.g., force line of conventional gooseneck or fifth wheel hitch). As illustrated in FIG. 1, angle α_1 is less than angle α_2 . By lowering the center line of the pulling force of tow vehicle **102**, angle α_1 is decreased to provide the more parallel, and in various embodiments, parallel pulling force of tow vehicle **102** on trailer **104**. Utilizing these concepts of trigonometry, geometry, and physics (e.g., lowering the force lines) improves the performance and lifespan of tow vehicle **102** and trailer **104**.

(21) The force line being more parallel to the road, and in various embodiments parallel to the road, decreases rocking of tow vehicle **102** (e.g., raising and lowering of the front and back of tow vehicle **102**) and trailer **104** (e.g., raising and lowering of the front and back of trailer **104**) as tow vehicle **102** and trailer **104** move over terrain that is not flat. Furthermore, lowering the center line of the pulling force may reduce the wear on tow vehicle **102** and trailer **104** by decreasing the amount of up and down (e.g., along the z-axis) and lateral (e.g., along the x-axis) movement of both tow vehicle **102** and trailer **104**. The up, down, and lateral movement of tow vehicle **102** may affect the steering, wheel alignment, and tire wear of tow vehicle **102** as the combined weight of

tow vehicle **102** and trailer **104** shifts from forward and backward on tow vehicle **102**. Therefore, lowering the centerline of the pulling force may reduce the amount and/or frequency of maintenance for tow vehicle **102** and/or trailer **104**. Additionally, lowering the center line of the pulling force may also increase the fuel efficiency of tow vehicle **102**.

(22) Referring now to FIG. 2, a trailer ball and receiver assembly **200** is illustrated, in accordance with various embodiments. Trailer ball and receiver assembly **200** may be an example of trailer ball and receiver system **112**, described above with respect to FIG. 1. Trailer ball and receiver assembly **200** includes a trailer ball **202**, a trailer coupler hitch **204**, and a receiver assembly **206**. Trailer ball **202** is spherical in shape. Receiver assembly **206** includes a socket **208** having a semi-spherical shape that is configured to receive trailer ball **202**. In the depicted embodiment, trailer ball and receiver assembly **200** does not use any type of lubrication between trailer ball **202** and receiver assembly **206**. This may be referred to as a dry socket or a dry fit. During use, trailer ball **202** abuts against socket **208** of receiver assembly **206** creating heat, wear, tear, and/or abrasion on the trailer ball and receiver assembly **200** (e.g., trailer ball **202** and receiver assembly **206**). This may result in a reduction in diameter, size, and/or various surface deformations of trailer ball **202** over time.

(23) The size of trailer ball **202** and socket **208** of receiver assembly **206** may be increased to mitigate the effects of a dry socket. In various embodiments, trailer ball **202** has a diameter **D1** that is about 2 5/16 inches (about 5.87 centimeters) to about 8 inches (about 20.3 centimeters), and more specifically, about 4 inches (about 10.2 centimeters) to about 7 inches (about 17.8 centimeters), and even more specifically, about 6 inches (about 15.2 centimeters). Socket **208** of receiver assembly **206** has a diameter **D2** that is about 1/16 inch (about 0.159 centimeter) to about 5/16 inches (about 0.794 centimeter) larger than diameter **D1**, and more specifically, about 3/16 inches (about 0.476 centimeter) larger than diameter **D1**. This configuration, in various embodiments, may reduce stress and/or strain on trailer ball and receiver assembly **200** by increasing the surface area over which the weight of a trailer (e.g., trailer **104**, FIG. 1) is distributed as compared to a conventional 2 5/16 inches (5.87 centimeters) trailer ball.

(24) Referring now to FIG. 3, a trailer ball and receiver assembly **300** is illustrated, in accordance with various embodiments. Trailer ball and receiver assembly **300** may be an example of trailer ball and receiver system **112** described above with respect to FIG. 1. Trailer ball and receiver assembly **300** includes similar components as those described above for trailer ball and receiver assembly **200** with respect to FIG. 2, including a trailer ball **302**, a trailer coupler **304**, a receiver assembly **306**, and a socket **308** of receiver assembly **306**. Descriptions of each component may not be repeated below. Trailer ball and receiver assembly **300** further includes one or more lubrication channels **310** and one or more grease fittings **312**, also referred to as grease Zerk.

(25) The one or more lubrication channels **310** extend from one or more inlets **314** at a top surface **306a** of receiver assembly **306** into socket **308** of receiver assembly **306**. That is, socket **308** of receiver assembly **306** is fluidly coupled with top surface **306a** of receiver assembly **306** by the one or more lubrication channels **310**. In various embodiments, the one or more lubrication channels **310** may be channels, tubes, or piping, among other structures, that allow a lubricant **316** to flow from outside of receiver assembly **306** into socket **308**. In various embodiments, lubricant **316** may be any type of lubricant, such as grease, dry lubricants, wet lubricants, oil-based, water-based, plant-based, and silicone-based, among others. In various embodiments, receiver assembly **306** may include a single lubrication channel **310**. In various embodiments, receiver assembly **306** may include more than two lubrication channels **310**. In various embodiments, each lubrication channel **310** may exit into socket **308** at a point that higher or lower (e.g., in the z-direction) than depicted in FIG. 3.

(26) In various embodiments, one or more of the lubrication channels **310** may further include a release valve at the outlet connected to socket **308**. In various embodiments, grease fitting **312** may be located at the inlet of each lubrication channel **310**. Grease fitting **312**, also referred to as a grease Zerk, provides a port through which lubricant **316** may be introduced into lubrication

channel **310**. Additionally, grease fitting **312** may prevent back flow of lubricant from lubrication channel **310**. In various embodiments, grease fitting **312** may be used to pressurize lubrication channel **310**, such as for injecting lubricant into receiver assembly **306** under pressure. For instance, a tool conventionally known as a grease gun may be attached to the grease fitting **312** and may inject lubricant **316** into lubrication channel **310** and socket **308** of receiver assembly **306** under pressure. In various embodiments, receiver assembly **306** may use an open/close valve at inlet **314** of each lubrication channel **310** to introduce lubricant into socket **308** instead of grease fitting **312**. In various embodiments, the open/close valve may be a one-way or a two-way valve. (27) Trailer ball and receiver assembly **300** may optionally include a locking collar **318** disposed on top surface **306a** and above socket **308** (e.g., the positive z-direction) and around trailer ball **302**. Locking collar **318** may be secured to top surface **306a** of receiver assembly **306** to prevent trailer ball **302** from unintentionally exiting socket **308**. In various embodiments, locking collar **318** may have an annular shape. In various embodiments, locking collar **318** may be coupled between trailer ball **302** and a trailer coupler hitch (e.g., coupler hitch **120**, FIG. 1) such that it is a part of the trailer (e.g., trailer **104**, FIG. 1).

(28) Referring now to FIG. 4, a trailer ball and receiver assembly **400** is illustrated, in accordance with various embodiments. Trailer ball and receiver assembly **400** may be an example of trailer ball and receiver system **112** described above with respect to FIG. 1. Trailer ball and receiver assembly **400** includes similar components as those described above with respect to trailer ball and receiver assembly **300** in FIG. 3, including a trailer ball **402**, a trailer coupler **404**, a receiver assembly **406**, a socket **408**, a lubrication channel **410**, a lubricant **416**, and an optional locking collar **418**. Descriptions of each component may not be repeated below. Trailer ball and receiver assembly **400** further includes one or more lubricant reservoirs **420** and one or more sensors **422**.

(29) The one or more lubricant reservoirs **420** may hold any type of lubricant **416**. In various embodiments, lubricant **416** may include grease, dry lubricants, wet lubricants, oil-based, water-based, plant-based, and silicone-based, among others. Each lubricant reservoir **420** may be connected to socket **408** by lubrication channel **410**, as described above for lubrication channels **310** (FIG. 3). Each lubricant reservoir **420** may include a lid, a hatch, or a valve, among other components, to secure lubricant **416** inside lubricant reservoir **420** and allow the filling of lubricant reservoir **420** with lubricant **416**. In various embodiments, lubricant **416** may be gravity fed to socket **408** from lubricant reservoir **420**.

(30) In various embodiments, trailer ball and receiver assembly **400** may further include a component **424** located within lubrication channel **410** to automatically feed, or release, lubricant **416** from lubricant reservoir **420** in response to an input. In various embodiments, component **424** may be a pump, a valve, or other mechanism that can open and close in response to the input, or signal. The one or more sensors **422** may include a temperature sensor, a viscosity sensor, and/or a fill sensor, among others. Each sensor **422** may be able to detect frictional heat of trailer ball and receiver assembly **400** and cause component **424** to provide lubricant **416** in response to the detected frictional heat. In various embodiments, sensor **422** may detect one or more other characteristic and may cause component **424** to provide lubricant **416** in response thereto. For instance, sensor **422** may be a motion sensor, timer, accelerometer, vibration sensor, or any other sensor or combination of sensors as desired. In various embodiments, sensor **422** may be disposed within trailer ball **402**. In various embodiments, sensor **422** may be disposed within receiver assembly **406**. In various embodiments, each sensor **422** may be configured to provide a signal to one or more components **424**. The signal may include an open signal and a close signal. In this way, dispensing lubricant **416** may be automated. Each sensor **422** may be set to release or hold lubricant **426** at a set threshold temperature and/or threshold viscosity or other characteristic or combination of characteristics. That is, when the threshold temperature and/or threshold viscosity are met, then component **424** may release lubricant **416** to be introduced to socket **408**.

Alternatively, sensor **422** may communicate with one or more computers (that have pre-

programmed threshold temperature and/or threshold viscosity) via a network that can further communicate with the one or more components **424** (e.g., pumps and/release valves, etc.) via a network to release lubricant **416** to socket **408**. The system may automatically cease feeding lubricant **416** to socket **408** in response to a preset, or lower threshold, temperature and/or viscosity being detected by sensor **422**. This process may be repeated continuously during operation of tow vehicle **102** and trailer **104**.

(31) Referring now to FIGS. 5A-5E, a receiver assembly **500** is illustrated, in accordance with various embodiments. In various embodiments, receiver assembly **500** may be an example of receiver assembly **144** described above with respect to FIG. 1. Receiver assembly **500** includes a ball receiver **506**, a cylinder **542**, and a plug **560**. In various embodiments, ball receiver **506** is coupled to cylinder **542** and plug **560** is coupled to ball receiver **506** (such as in FIG. 5D). FIG. 5A illustrates a perspective view from the top of ball receiver **506**, FIG. 5B illustrates a perspective view from the bottom of ball receiver **506**, FIG. 5C illustrates a top down view of ball receiver **506**, FIG. 5D illustrates a side view of receiver assembly **500** showing both ball receiver **506** and cylinder **542** connected together, and FIG. 5E illustrates a side view of plug **560**.

(32) Ball receiver **506** includes an outer sidewall **530**, a socket **532**, a bottom surface **534**, a top surface **536**, and a lubricant channel **538**. Ball receiver **506** may be generally cylindrical in shape with outer sidewall **530** being vertical (e.g., parallel to the z-axis). Outer sidewall **530** extends from top surface **536** to bottom surface **534**. Top surface **536** and bottom surface **534** may be parallel to each other. In various embodiments, bottom surface **534** is flat (e.g., parallel to the x-y-plane). Socket **532** is disposed in ball receiver **506** and has a semi-spherical shape that is open upward (e.g., in the positive z-direction). Socket **532** is configured to receive a spherical shaped trailer ball (e.g., trailer ball **140**, trailer ball **202**, trailer ball **302**, etc.) In various embodiments, socket **532** is open toward a bed of the truck (e.g., bed **108** of tow vehicle **102**). Ball receiver **506** has an outer diameter **D3** that is about 5 inches (about 12.7 centimeters) to about 8 inches (about 20.3 centimeters), and more specifically, about 6.5 inches (about 16.5 centimeters). Socket **532** has a diameter **D4** that may be alternatively referred as an inner diameter **D4** of ball receiver **506**. Diameter **D4** is about 5.5 inches (about 14 centimeters) to about 7.5 inches (about 19.1 centimeters), and more specifically, about 6.125 inches (about 15.6 centimeters). Ball receiver **506** has a height **H1** that is about 6 inches (about 15.2 centimeters) to about 8 inches (about 20.3 centimeters), and more specifically, about 6.5 inches (about 16.5 centimeters).

(33) Lubricant channel **538** may be located at the bottom of ball receiver **506** (e.g., the negative z-direction) allowing liquid to be drained from ball receiver **506**, and more specifically, drained from socket **532**. Lubricant channel **538** has a length **L1** extending from a bottom of socket **532** (e.g., the negative z-direction) to bottom surface **534** of ball receiver **506** (e.g., the negative z-direction). Length **L1** may be about 0.5 inches (about 1.27 centimeters) to about 2 inches (about 5.08 centimeters), and more specifically, about 1 inch (about 2.54 centimeters). Lubricant channel **538** is configured to receive plug **560** to retain liquid (e.g., a lubricant such as described above with respect to FIGS. 3 and 4). In various embodiments, lubricant channel **538** may include threads **539** that are configured to receive and secure plug **560**. Plug **560** may include complementary threads **562** configured to engage threads **539** of lubricant channel **538**. Plug **560** may further include a head **564** that may be engaged by a tool (e.g., a wrench) to tighten and/or untighten plug **560** from lubricant channel **538**. Threads **562** of plug **560** has a length **L2** sufficient to engage and seal lubricant channel **538**. In various embodiments, length **L2** may be about 0.5 inches (about 1.27 centimeters) to about 2 inches (about 5.08 centimeters), and more specifically, about 1 inch (about 2.54 centimeters).

(34) In various embodiments, plug **560** may be press fit into lubricant channel **538**. In various embodiments, plug **560** may be a lever-operated plug that is secured in lubricant channel **538** by a lever. In various embodiments, plug **560** may be a twist-lock plug, locking into place in lubricant channel **538** using a twist mechanism.

(35) Referring now to FIG. 5D, ball receiver **506** may be coupled to cylinder **542** prior to, or as part of, installation in a tow vehicle (e.g., tow vehicle **102**, FIG. 1), to form receiver assembly **500**. In various embodiments, ball receiver **506** and cylinder **542** may be formed as a single monolithic component. Cylinder **542** has an outer diameter D5 and an inner diameter D6. Outer diameter D5 may be about equal to outer diameter D1 of ball receiver **506** and inner diameter D6 may be about equal to inner diameter D2 of ball receiver **506**. Cylinder **542** has a height H2 that may be adjusted for proper installation in tow vehicle (e.g., tow vehicle **102**, FIG. 1). Height H2 may be about 3 inches (about 7.62 centimeters) to about 36 inches (about 91.4 centimeters), and more specifically, about 6 inches (about 15.2 centimeters) to about 24 inches (about 61 centimeters). In various embodiments, ball receiver **506** may be welded to cylinder **542** with socket **532** opening upward (e.g., the positive z-direction) into cylinder **542**. Receiver assembly **500** (e.g., ball receiver **506** and cylinder **542**) has a height H3 that is about 9 inches (about 22.9 centimeters) to about 42 inches (about 107 centimeters), and more specifically, about 12 inches (about 30.5 centimeters) to about 30.5 inches (about 77.5 centimeters). In various embodiments, receiver assembly **500** may be cut to a height shorter than height H3 prior to installation in the tow vehicle. In various embodiments, cylinder **542** may be cut to a shorter height than height H2 prior to coupling cylinder **542** to ball receiver **506**.

(36) Ball receiver **506**, and more specifically, socket **532** is configured to receive a lubricant such as grease, oil, or other lubricants. In various embodiments, the lubricant may be lubricant **416** described above with respect to FIG. 4. The lubricant may be manually added to ball receiver **506** through the top of cylinder **542** (e.g., the positive z-direction). The lubricant is held in socket **532** by plug **560**. Plug **560** may be removed to drain the lubricant through lubricant channel **538**.

(37) Referring now to FIGS. 6A-6D, a trailer ball assembly **600** and its various components are illustrated, in accordance with various embodiments. Trailer ball assembly **600** includes a trailer ball **602** and a trailer coupler **604**. Trailer ball **602** includes a spherical ball portion **606**, a ball coupler portion **608**, and a securing hole **610** formed through ball coupler portion **608**. In various embodiments, ball coupler portion **608** may be cylindrical, rectangular, triangular, or hexagonal, among other shapes. Spherical ball portion **602** is configured to engage socket **538** of receiver assembly **506** as described above in FIGS. 5A-5E. Spherical ball portion **606** has a diameter D7 that is about 5 inches (about 12.7 centimeters) to about 7 inches (about 17.8 centimeters), and more specifically, about 6 inches (about 15.2 centimeters). In various embodiments, diameter D7 may be 8 inches or greater depending on design use, load requirements, trailer rotation forces (e.g., side winds, grade angles, shifting loads, etc. or a combination thereof). Diameter D7 may be about 1/16 inches (about 0.159 centimeter) to about 5/16 inches (about 0.794 centimeter) smaller than diameter D3, and more specifically, about 2/16 inches (about 0.318 centimeter) to about 3/16 inches (about 0.476 centimeter) smaller than diameter D3. FIGS. 6A and 6B illustrate perspective views of trailer ball **602**, FIG. 6C illustrates trailer coupler **604**, and FIG. 6D illustrates trailer ball assembly **600**.

(38) Ball coupler portion **608** is coupled to the top (e.g., the positive z-direction) of the spherical ball portion **606**. Ball coupler portion **608** has a diameter D8 and a length L3. Diameter D8 is about 2 inches (about 5.08 centimeters) to about 6 inches (about 15.2 centimeters), and more specifically, about 3 inches (about 7.62 centimeters) to about 4 inches (about 10.2 centimeters). Length L3 is about 2 inches (about 5.08 centimeters) to about 5 inches (about 12.7 centimeters), and more specifically, about 3 inches (about 7.62 centimeters) to about 4 inches (about 10.2 centimeters). Securing hole **610** is formed near the top (e.g., the positive z-direction) of ball coupler portion **608** and extends through ball coupler portion **608**.

(39) Trailer coupler **604** includes a trailer coupler adapter **612**, a trailer adapter plate **614**, a damper **616**, a ball mount **618**, and a ball mount plate **620**. Trailer coupler **604** is configured to interface with a trailer (e.g., trailer **104**), and more specifically, with a coupler hitch (e.g., coupler hitch **120**). In various embodiments, the trailer may be a gooseneck trailer. Trailer coupler **604** may replace the

existing coupler hitch allowing any trailer to be converted to use the trailer ball and receiver assembly (e.g., trailer ball and receiver system **112**, FIG. **1**) disclosed herein with minimal changes to the trailer.

(40) Ball mount **618** is configured to receive and secure trailer ball **602**. Ball mount **618** has a diameter **D9** and a length **L4**. Diameter **D9** is about 3 inches (about 7.62 centimeters) to about 7 inches (about 17.8 centimeters), and more specifically, about 4 inches (about 10.2 centimeters) to about 5 inches (about 12.7 centimeters). Length **L4** is about 6 inches (about 15.2 centimeters) to about 24 inches (about 61 centimeters), and more specifically, about 9 inches (about 22.9 centimeters) to about 18 inches (about 45.7 centimeters). Ball mount **618** includes a pair of holes **628** configured to align with securing hole **610** of trailer ball **602**. Ball mount **618** further includes a pair of holes **628** formed therethrough that may be used to engage and secure trailer ball **602**.

(41) Ball mount plate **620** is coupled to a top portion of ball mount **618** (e.g., the positive z-direction). In various embodiments, ball mount plate **620** is welded to ball mount **618**. Ball mount plate **620** has a diameter **D10** that is about 5 inches (about 12.7 centimeters) to about 8 inches (about 20.3 centimeters), and more specifically, about 5.5 inches (about 14 centimeters) to about 6.5 inches (about 16.5 centimeters). In various embodiments, ball mount plate **620** is sized to fit inside receiver assembly **500**, and more specifically, inside cylinder **542**.

(42) Trailer coupler adapter **612** has a diameter **D11** and a length **L5**. Diameter **D11** is about 3 inches (about 7.62 centimeters) to about 7 inches (about 17.8 centimeters), and more specifically, about 4 inches (about 10.2 centimeters) to about 5 inches (about 12.7 centimeters). Length **L5** is about 6 inches (about 15.2 centimeters) to about 48 inches (about 122 centimeters), and more specifically, about 18 inches (about 45.7 centimeters) to about 36 inches (about 91.4 centimeters). Trailer coupler adapter **612** includes a plurality of holes **622** for along length **L5**. The plurality of holes **622** are configured to secure trailer ball assembly **600**, and more specifically, trailer coupler adapter **612** to the trailer (e.g., trailer **104**, FIG. **1**). In various embodiments, the plurality of holes **622** are arranged in a straight line along length **L5**. In various embodiments, such as the illustrated embodiment, the plurality of holes **622** are aligned so that alternating holes are laterally offset from the vertically adjacent holes.

(43) Trailer adapter plate **614** is coupled to a bottom portion of trailer coupler adapter **612** (e.g., the negative z-direction). In various embodiments, trailer adapter plate **614** is welded to trailer coupler adapter **612**. Trailer adapter plate **614** has a diameter **D12** that is about 5 inches (about 12.7 centimeters) to about 8 inches (about 20.3 centimeters), and more specifically, about 5.5 inches (about 14 centimeters) to about 6.5 inches (about 16.5 centimeters). In various embodiments, trailer adapter plate **614** is sized to fit inside receiver assembly **500** (FIG. **5D**), and more specifically, inside cylinder **542** (FIG. **5D**).

(44) Damper **616** is disposed between, and coupled to, trailer adapter plate **614** and ball mount plate **620** when trailer ball assembly **600** is assembled. Damper **616** has a diameter **D13** and a height **H4**. Diameter **D13** is about 5 inches (about 12.7 centimeters) to about 8 inches (about 20.3 centimeters), and more specifically, about 5.5 inches (about 14 centimeters) to about 6.5 inches (about 16.5 centimeters). Height **H4** is about 2 inches (about 5.08 centimeters) to about 6 inches (about 15.2 centimeters), and more specifically, about 3 inches (about 7.62 centimeters) to about 4 inches (about 10.2 centimeters). Trailer adapter plate **614**, damper **616**, and ball mount plate **620** may be coupled together using a plurality of bolts **624**, nuts **626**, and, in various embodiments, washers. Damper **616** may include a plurality of holes through which bolts **624** may extend.

(45) Trailer ball assembly **600** further includes a pin **630** and cotter pin **632** that may be used to secure trailer ball **602** to ball mount **618**. Trailer ball **602**, and more specifically, ball coupler portion **608** is inserted into ball mount **618**. Securing hole **610** and holes **628** are then aligned so that pin **630** may pass through securing hole **610** and holes **628**, thereby supporting trailer ball **602**. Pin **630** may include a handle, or other protrusion, at a proximal end of pin **630** to prevent pin **630** from passing entirely through securing hole **610** and holes **628**. Cotter pin **632** is inserted through a

distal end of pin 630, preventing pin 630 from being pulled back out of securing hole 610 and holes 628.

(46) Benefits, other advantages, and solutions to problems have been described herein with regard to specific embodiments. Furthermore, the connecting lines shown in the various figures contained herein are intended to represent exemplary functional relationships and/or physical couplings between the various elements. It should be noted that many alternative or additional functional relationships or physical connections may be present in a practical system. However, the benefits, advantages, solutions to problems, and any elements that may cause any benefit, advantage, or solution to occur or become more pronounced are not to be construed as critical, required, or essential features or elements of the disclosure. The scope of the disclosure is accordingly to be limited by nothing other than the appended claims, in which reference to an element in the singular is not intended to mean “one and only one” unless explicitly so stated, but rather “one or more.” Moreover, where a phrase similar to “at least one of A, B, or C” is used in the claims, it is intended that the phrase be interpreted to mean that A alone may be present in an embodiment, B alone may be present in an embodiment, C alone may be present in an embodiment, or that any combination of the elements A, B and C may be present in a single embodiment; for example, A and B, A and C, B and C, or A and B and C. Different cross-hatching is used throughout the figures to denote different parts but not necessarily to denote the same or different materials.

(47) Systems, methods and apparatus are provided herein. In the detailed description herein, references to “one embodiment,” “an embodiment,” “various embodiments,” etc., indicate that the embodiment described may include a particular feature, structure, or characteristic, but every embodiment may not necessarily include the particular feature, structure, or characteristic. Moreover, such phrases are not necessarily referring to the same embodiment. Further, when a particular feature, structure, or characteristic is described in connection with an embodiment, it is submitted that it is within the knowledge of one skilled in the art to affect such feature, structure, or characteristic in connection with other embodiments whether or not explicitly described. After reading the description, it will be apparent to one skilled in the relevant art(s) how to implement the disclosure in alternative embodiments.

(48) Numbers, percentages, or other values stated herein are intended to include that value, and also other values that are about or approximately equal to the stated value, as would be appreciated by one of ordinary skill in the art encompassed by various embodiments of the present disclosure. A stated value should therefore be interpreted broadly enough to encompass values that are at least close enough to the stated value to perform a desired function or achieve a desired result. The stated values include at least the variation to be expected in a suitable industrial process, and may include values that are within 5% of a stated value. Additionally, the terms “substantially,” “about” or “approximately” as used herein represent an amount close to the stated amount that still performs a desired function or achieves a desired result. For example, the term “substantially,” “about” or “approximately” may refer to an amount that is within 5% of a stated amount or value.

(49) Furthermore, no element, component, or method step in the present disclosure is intended to be dedicated to the public regardless of whether the element, component, or method step is explicitly recited in the claims. No claim element herein is to be construed under the provisions of 35 U.S.C. 112(f) unless the element is expressly recited using the phrase “means for.” As used herein, the terms “comprises,” “comprising,” or any other variation thereof, are intended to cover a non-exclusive inclusion, such that a process, method, article, or apparatus that comprises a list of elements does not include only those elements but may include other elements not expressly listed or inherent to such process, method, article, or apparatus.

(50) Finally, it should be understood that any of the above-described concepts can be used alone or in combination with any or all of the other above-described concepts. Although various embodiments have been disclosed and described, one of ordinary skill in this art would recognize that certain modifications would come within the scope of this disclosure. Accordingly, the

description is not intended to be exhaustive or to limit the principles described or illustrated herein to any precise form. Many modifications and variations are possible in light of the above teaching.

Claims

1. A system, comprising: a tow vehicle including: a bed having a first surface and an opposing second surface; and a receiver assembly coupled to the bed, the receiver assembly extending through the first surface and the second surface, the receiver assembly including: a hollow cylinder having a first end and a second end, the first end of the hollow cylinder coupled to the bed of the tow vehicle; and a ball receiver having a semi-spherical shape, the ball receiver coupled to the second end of the hollow cylinder; and a trailer including: a deck; a neck coupled to the deck; and a trailer coupler coupled to the neck, the trailer coupler including: a trailer ball configured to engage the receiver assembly of the tow vehicle.
 2. The system of claim 1, further comprising: a lubricant disposed in the receiver assembly, the lubricant disposed between the trailer ball and the ball receiver.
 3. The system of claim 2, the receiver assembly further comprising: a channel formed in and through a bottom of the ball receiver, the channel configured to drain the lubricant from the ball receiver; and a removable plug configured to be inserted into the channel, the removable plug preventing the lubricant from draining from the ball receiver.
 4. The system of claim 3, the trailer coupler further comprising: a trailer coupler adapter; a ball mount configured to receive the trailer ball; and a damper disposed between the ball mount and the trailer coupler adapter.
 5. The system of claim 4, further comprising: a plurality of holes formed along a length of the trailer coupler adapter, the plurality of holes configured to secure the trailer coupler adapter to the neck of the trailer.
 6. The system of claim 1, wherein the trailer ball has a first diameter, the first diameter being about 5 inches to about 7 inches.
 7. The system of claim 6, wherein the semi-spherical shape of the ball receiver has a second diameter, the second diameter being about 1/16 inch larger than the first diameter.
 8. A trailer ball and receiver assembly, comprising: a trailer ball assembly coupled to a trailer, the trailer ball assembly including a trailer ball; and a ball receiver assembly coupled to a bed of a tow vehicle and extending below the bed of the tow vehicle, the ball receiver assembly including a semi-spherical socket configured to receive the trailer ball assembly, the receiver assembly further comprising: a cylinder having a first end and opposing second; and a hole formed from the first end to the second end, wherein the second end of the cylinder is coupled to the ball receiver assembly so that the semi-spherical socket opens toward the hole.
 9. A trailer ball and receiver assembly, comprising: a trailer ball assembly coupled to a trailer, the trailer ball assembly including a trailer ball; a ball receiver assembly coupled to a bed of a tow vehicle and extending below the bed of the tow vehicle, the ball receiver assembly including a semi-spherical socket configured to receive the trailer ball assembly; a ball mount configured to receive the trailer ball and secure the trailer ball in place; and a trailer coupler adapter having a first end and a second end, the first end coupled to the ball mount and the second end coupled to the trailer.
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